CENG 111
Introduction to Computer Engineering Concepts
2026-2027 Fall

Instructors:

Dr. Hüseyin Aydın, Dr. Sinan Kalkan

Teaching Assistants:

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Course Schedule:

Section 1	Section 2
Tue: 11:40, 12:40 (BMB-1) Thu: 11:40 (BMB-1)	Tue: 9:40, 10:40 (BMB-1) Thu: 10:40 (BMB-1)

Lab Schedule: TBA. **Prerequisites:** None

Catalog Description:

Introduction to the fundamentals of computer systems, including computer organization, operating systems, language processors and user interfaces. Introduction to algorithms and programming. Reasoning informally about the correctness and efficiency of programs. A functional programming language will be used for practical work.

Course Objectives:

To provide a basic understanding of fundamental concepts in computer science and engineering. To improve the skills to work with abstract notions for solving computational problems. Teaching a particular programming language is not a primary objective; the language will serve as a medium for experimentation.

Textbooks:

- *Introduction to Computer Hardware and Systems: Lecture Notes,* by G. Üçoluk and S. Kalkan, 2024.
- *Introduction to Programming Concepts with Case Studies in Python,* by G. Üçoluk and S. Kalkan, Springer 2012.

Outline:

Week	*Notes*	*Course Material*	*Lab Cov.*
Week1 28.09		Introduction [Course content, objectives, outline; Grading; Information about the homeworks, the labs, the exams; Newsgroup, Website] Computing and Computation [What is computing/computation? Digital computation, analog computation and nature's computation: Pros and Cons.]	No lab
Week2 05.10		Computing and Computation [continued] [Digital computation; Basic digital computational elements: Switches, transistors; How to make basic gates; How to build bigger/complex circuits; Boolean Logic – Digital Circuit connection; Truth table; 1-bit addition, N-bit addition; Memory]	No lab
Week3 12.10		Von Neumann Architecture [The separation abstraction of Memory, CPU; Fetch-Decode-Execute cycle; Machine-level instructions; Main registers: MAR, MDR, IR, PC; Comparison with the Harvard Architecture; Pros and Cons of separating data and instructions] Computer as a Turing machine [A short discussion on what a Turing machine is]	Department's Electronic Services, Newsgroups, Introduction to Linux
Week4 19.10		Introduction to Python [An overview of the main features and version differences] Basic Data Types [Binary representation of integers, real numbers, characters, Boolean values and strings and the corresponding data types in Python]	Linux Commands
Week5 26.10	29th is holiday	Container Data Types [array vs. non-array containers; strings, tuples and lists in Python; operators with container data types in Python; type conversion in Python]	Linux Commands

Week6 02.11		Expression Evaluation [Expressions and their evaluation; operators' precedence and evaluation; prefix, infix, postfix expressions; an overview of expression evaluation algorithm] Actions for Input-Output [Different mechanisms for I/O in Python]	Quiz
Week7 09.11		Basic and Compound Statements [single and multiple assignment; del and other statements; compound statements] Variables, Assignment and Aliasing Conditional Execution [if statements; nested if statements; conditional expression]	Python introduction, Basic data types
Week8 16.11		Functions [Defining a function, value of functions, call by value, sharing and reference]	Advanced data types (str, list, tuple)
Week9 23.11		Recursion	Functions and conditionals
Week10 30.11		MIDTERM	Recursion
Week11 07.12		Iteration [for and while statements; list/set comprehension]	Iteration: for, while statements
Week12 14.12		Abstract Data Types [Stack; Queue; Priority Queue]	Iteration and Recursion - Recap
Week13 21.12		Abstract Data Types [Trees; Binary (Search) Trees]	Abstract Data Types: Stack, Queue and Tree
Week14 28.12		Object-oriented Programming [Encapsulation, inheritance, polymorphism, etc. Examples with OOP.] Overview of the world of programming [Compiled vs interpreted languages; Programming paradigms: Imperative, Functional, Object-oriented, Declarative-Logical, Parallel-Concurrent; How do we choose among the paradigms and the available languages; Current trend.]	More on trees: N-ary Trees

Grading:

Midterm	%27
Lab Exams (3-4)	%30
Von Neumann Machine Exam	%5
Participation	%3
Final	%35

- Most assignments will be programming assignments (some could be exercises to be written up).
- Laboratory schedule will be announced later. Keep watching.
- Starting with the Weeks 4-5 the laboratory sessions will go in parallel with the lectures. You will be asked to solve problems by using the ideas introduced in the lectures of that week. Earlier lab sessions will build your computing/communicating literacy.
- Attending the lab sessions is mandatory. Missing two lab sessions will result in a grade of NA for the course.

Web:

The webpage of the course will contain various material regarding the course: http://ceng.metu.edu.tr/ceng111

ODTUclass:

All communication, announcements, assignment collection etc. will take place at https://odtuclass.ceng.metu.edu.tr/

You **should** check this page *every day*.

METU Code of Ethics and Core Values:

"The members of the METU community are reliable, responsible and honourable people who embrace only the success and recognition they deserve, and act with integrity in their use, evaluation and presentation of facts, data and documents."

All students are expected to honour this code and the core values outlined at: https://www.metu.edu.tr/code-ethics-core-values